

Comparative Analysis of Sampling-Based Motion Planning in Constrained Environments

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ABSTRACT

This poster evaluates sampling-based motion planning algorithms (RRT, RRT*, PRM, and an custom Adaptive RRT) in a structured maze environments. We compare planners using planning time, path length, and a smoothness metric based on angular deviation. Highlighting trade-offs between speed and path quality.

INTRODUCTION

Motion planning has always been a fundamental problem in robotics that aims to compute a collision-free path from a start configuration x_{start} to a goal configuration x_{goal} within a constrained environment. In continuous spaces, this becomes a challenging problem due to high dimensionality and complex obstacle geometry.

To tackle this problem, sampling-based motion planners achieve this by avoiding explicitly construction of the configuration space. Instead, they generate random samples in the free space X_{free} and build structures such as trees or graphs to explore feasible paths.

The most widely used sampling-based planners are:

- **Rapidly Exploring Random Tree (RRT)**, which incrementally expands a tree toward randomly sampled states, enabling fast discovery of feasible paths.
- **RRT*** an extension of RRT that improves path quality through a rewiring process, achieving asymptotic optimality at increased computational cost.
- **Probabilistic Roadmap (PRM)**, which constructs a graph of sampled configurations and connects nearby nodes, enabling efficient path queries through graph search.

With each planner achieving different results with different trade-offs between computational efficiency and path quality, particularly for extra complex constraints.

PROJECT OBJECTIVE

In this poster, we execute a comparative evaluation of sampling-based motion planners in a structured maze environment. Specifically, we analyze using three key metrics

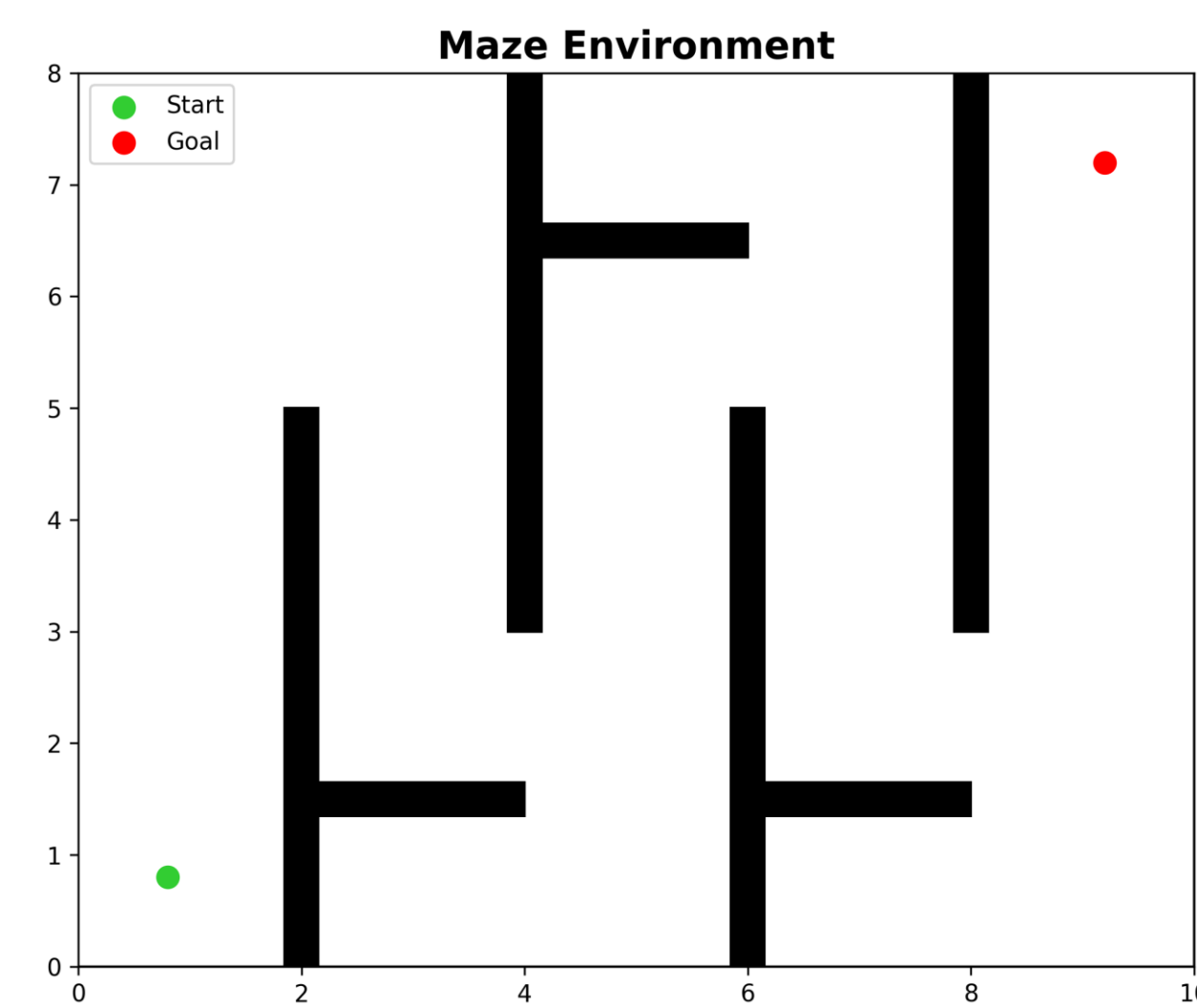
- Planning Time
- Path Length
- Path Smoothness

Additionally, we also introduced an **adaptive goal-biased RRT variant**, which aims to increase the probability of sampling toward the goal to improve convergence efficiency.

Our goal is to characterize the trade-offs between speed, optimality and trajectory quality across different planning algorithms.

METHODOLOGY

We evaluate the sampling-based planners in a 2D maze environment with rectangular obstacles. The configuration space is defined as $X \subset \mathbb{R}^2$ and valid states lie in the collision-free region X_{free}



A feasible path is defined as a continuous function

$$\sigma: [0, 1] \rightarrow X_{free}, \quad \sigma(0) = x_{start}, \quad \sigma(1) = x_{goal}$$

Planners Evaluated:

- RRT: Expands a tree by sampling random states and extending towards them:

$$x_{new} = x_{nearest} + \eta \cdot \frac{x_{rand} - x_{nearest}}{|x_{rand} - x_{nearest}|}$$

- RRT*: Improves path quality via rewiring using a cost function:

$$c(x) = \{\text{cost} - \text{to} - \text{come}\},$$

$$x_{parent} = \arg \min_{x \in \mathcal{N}} [c(x) + |x - x_{new}|]$$

- PRM: builds a roadmap graph from sampled states

$$V = \{x_1, \dots, x_n\}, \quad E = \{(x_i, x_j) \mid |x_i - x_j| < r\}$$

- Adaptive Goal-Biased RRT: Samples the goal with probability p to accelerate convergence

$$x_{rand} = \begin{cases} x_{goal} & \text{with probability } p \\ \text{random sample} & \text{otherwise} \end{cases}$$

Each planner is executed over 30 trials and performance is evaluated using the following metrics:

1. Planning Time: The time it takes to compute its final path

$$T = t_{end} - t_{start}$$

2. Path Length: Total Euclidean distance along the path

$$L = \sum_{i=1}^{N-1} |x_{i+1} - x_i|$$

3. Path Smoothness: cumulative angular deviation between consecutive path segments

$$S = \sum_{i=1}^{N-2} |\theta_{i+1} - \theta_i| \text{ where } \theta_i = \tan^{-1} \left(\frac{y_{i+1} - y_i}{x_{i+1} - x_i} \right)$$

RESULTS

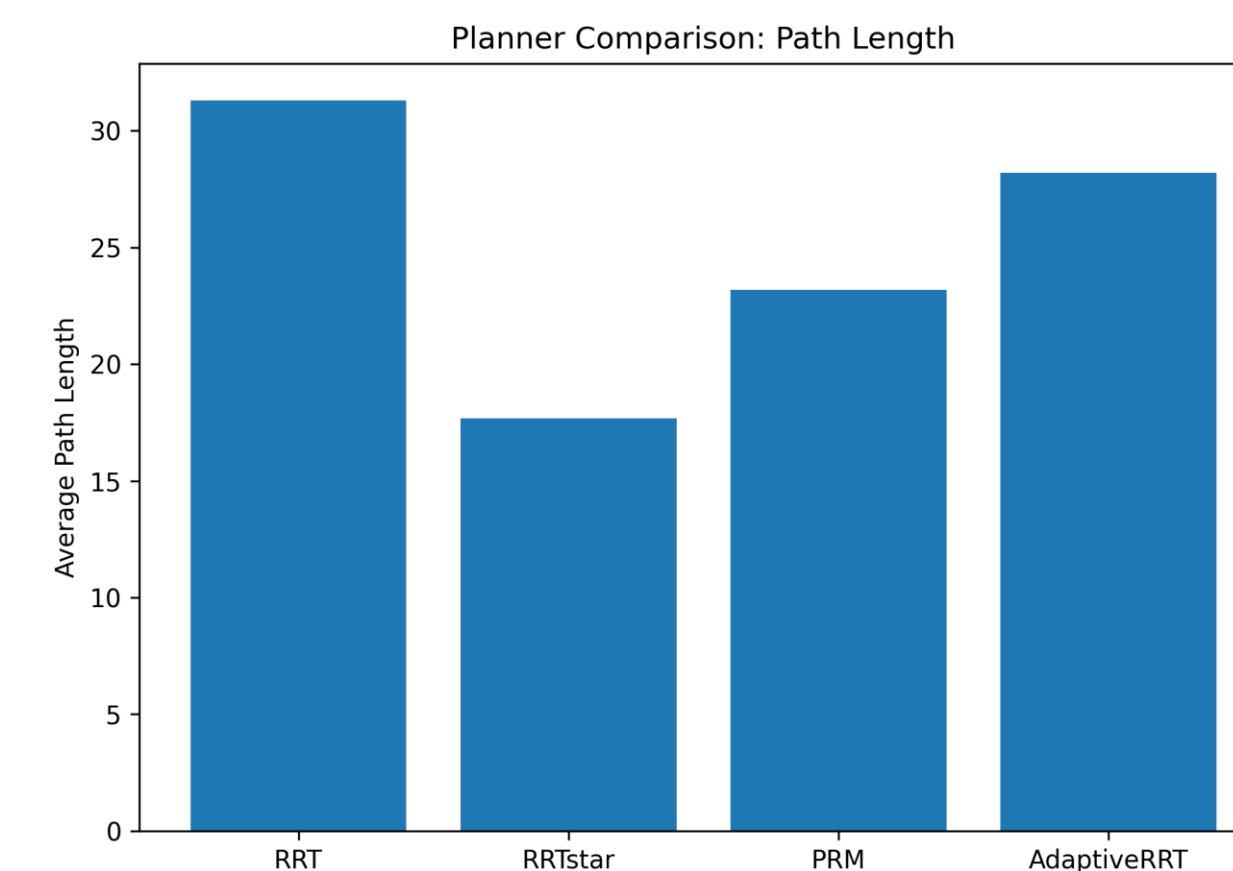
Qualitative Assessment:

Planning Time

- **RRT** was the fastest planner (5.3ms) due to its simple greedy expansion
- **Adaptive RRT** achieved competitive performance (~7.0 ms), slightly slower due to adaptive sampling overhead.
- **PRM** required more time (~11.1 ms) due to roadmap construction.
- **RRT*** was **orders of magnitude slower (~30,000 ms)** because of rewiring and optimization.

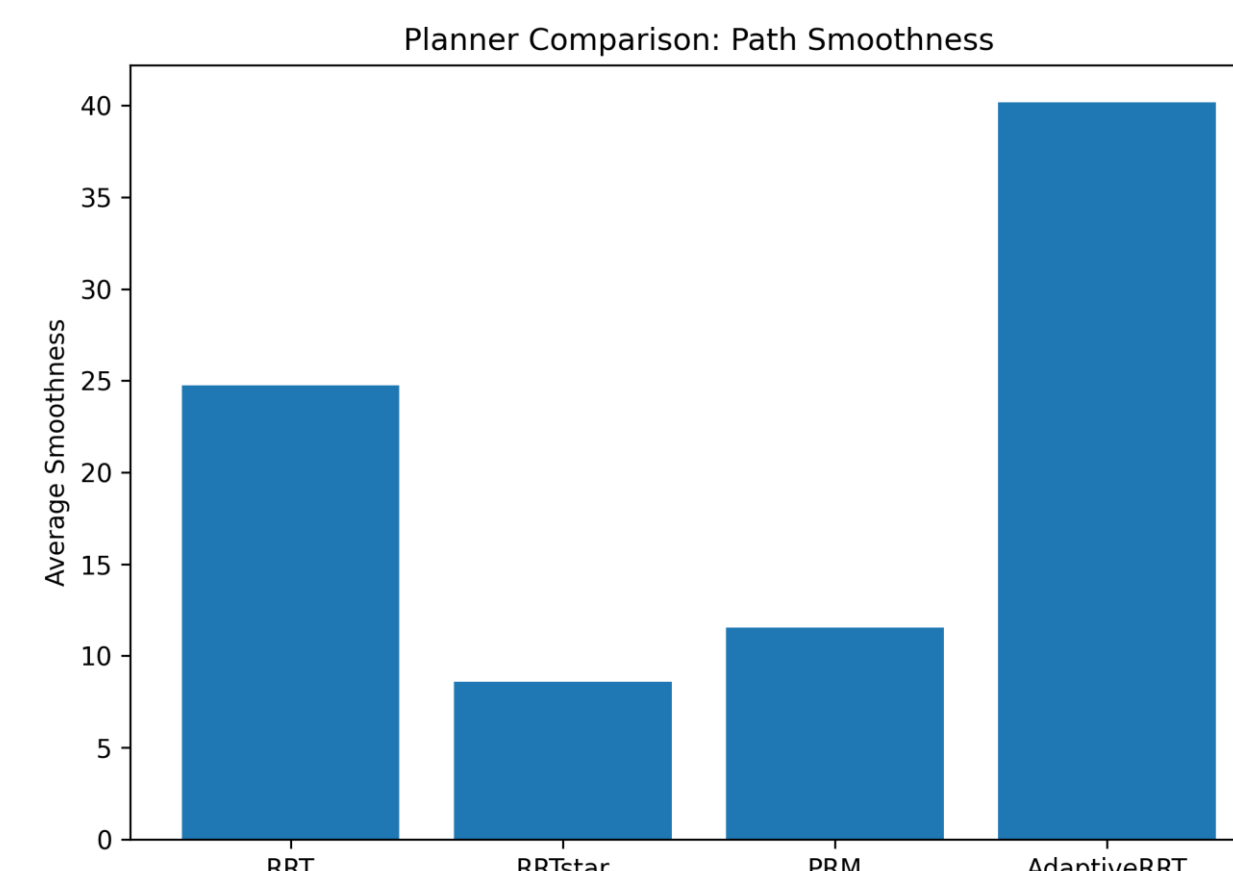
Path Length

- **RRT*** produced the **shortest paths (~17.6)** due to asymptotic optimality.
- **PRM (~23.3)** and **Adaptive RRT (~28.2)** achieved moderate improvements..
- **RRT (~31.3)** generated the longest, suboptimal paths.



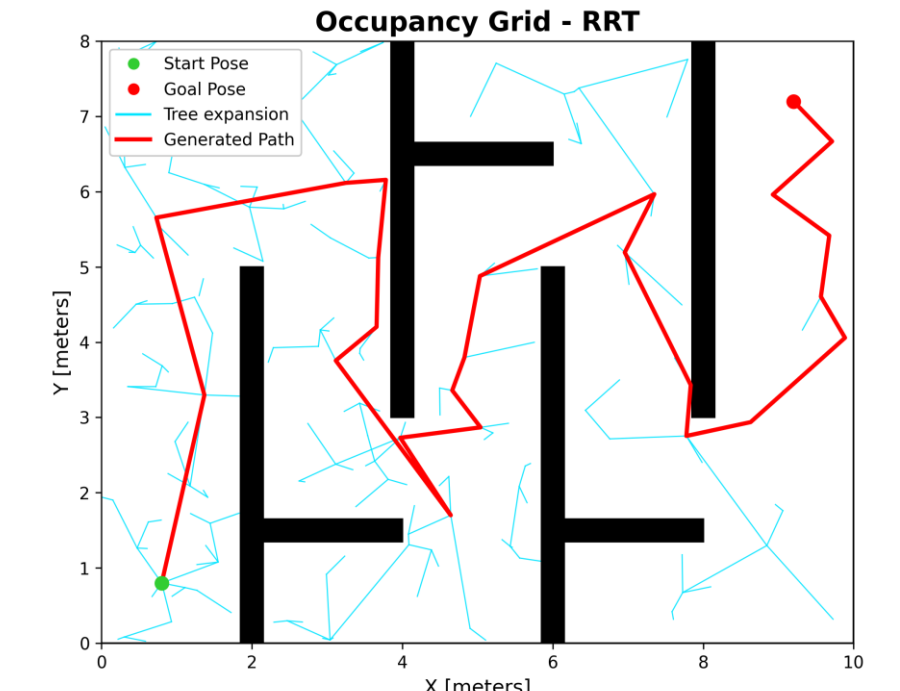
Path Smoothness

- **RRT*** produced the smoothest paths (~8.6).
- **PRM (~11.6)** also performed well due to global connectivity.
- **RRT (~24.7)** had jagged paths due to random expansion.
- **Adaptive RRT (~40.1)** showed the lowest smoothness due to aggressive exploration.

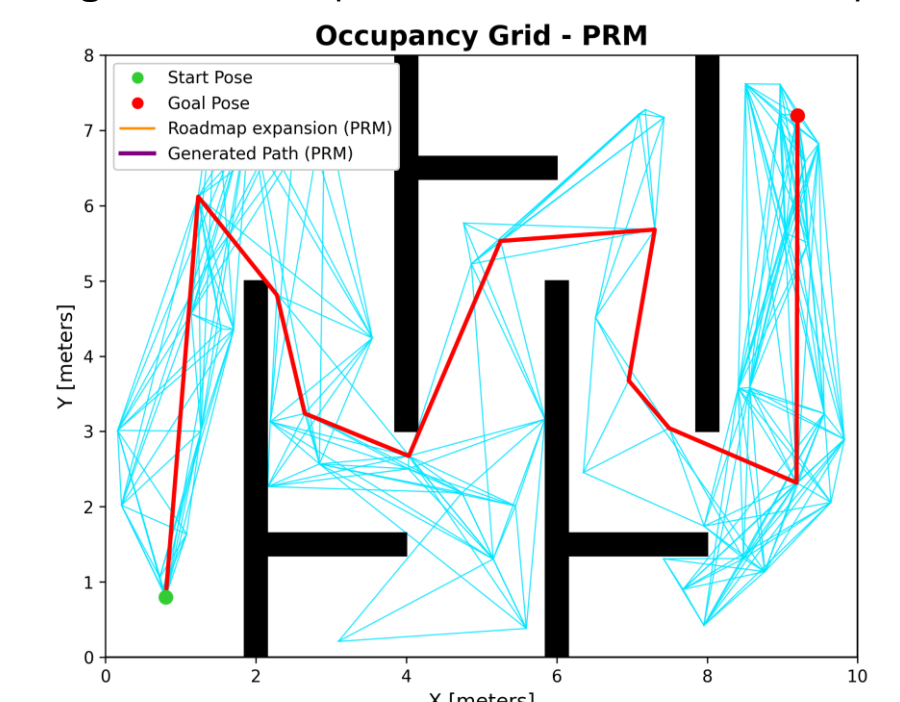


VISUAL COMPARISONS

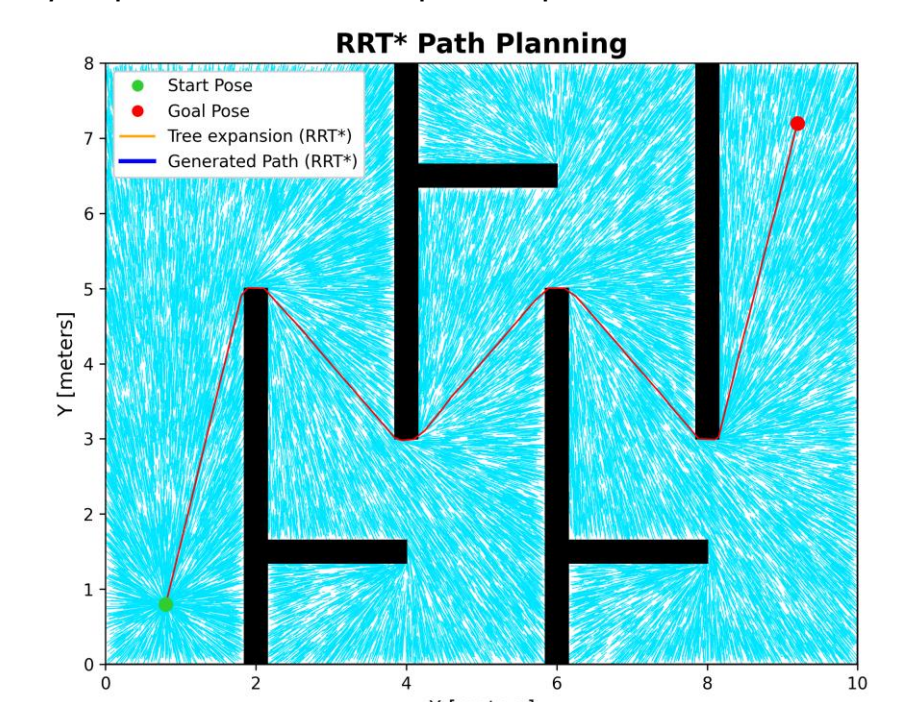
- **RRT** produces fast but irregular paths



- **PRM** builds a global roadmap with structured connectivity



- **RRT*** heavily explores to refine optimal paths



- **Adaptive RRT** balances exploration and goal direction but sacrifices smoothness

